Retrieval Quality Across Context Lengths

A controlled comparison of dense and reranked pipelines

Abstract

Longer retrieval contexts can improve coverage while increasing the work required to find useful evidence. We compare a dense retriever with a reranked pipeline at three context budgets on a synthetic evaluation set. The reranked pipeline achieves higher answer accuracy at every tested budget, with a smaller gain between 4k and 8k tokens. These observations describe this controlled example; they do not establish that more context improves every retrieval task.

1. Evaluation setup

We evaluate 1,200 questions using the same corpus, generator, and prompt for both pipelines. Context budgets are 2k, 4k, and 8k tokens. Each configuration is repeated with five random seeds. Accuracy is the fraction of answers matching the reference response; latency is measured on the same hardware. The training objective combines a task loss, a consistency term, and weight regularization:

$$\mathcal{J}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left[\alpha\,\mathcal{L}_{\text{task}}(y_i, f_\theta(x_i)) + \beta\,\mathcal{L}_{\text{cons}}(f_\theta(x_i), f_\theta(x_i^{'}))\right] + \lambda\lVert\theta\rVert_2^2 \tag{1}$$

Here N is the number of training examples; alpha, beta, and lambda are fixed across all configurations.

2. Results

Figure 1 compares accuracy across the three context budgets. The reranked pipeline rises from 68.4% at 2k to 76.1% at 8k, while the dense retriever rises from 64.2% to 70.0%. The largest observed gain occurs between 2k and 4k tokens for both pipelines. Error bars show one standard deviation across five seeds. These differences summarize the tested configurations and should not be interpreted as a causal effect of context length alone.

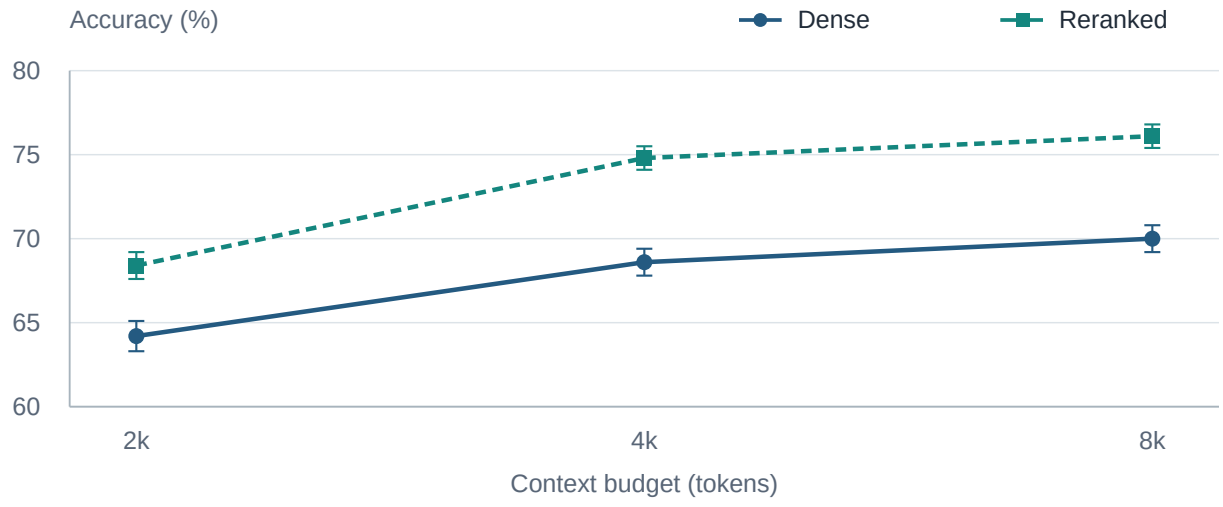

Figure 1. Answer accuracy by context budget. Points show the mean of five seeds; error bars show one standard deviation.

Table 1. Measurements at 4k tokens. Accuracy is mean +/- standard deviation over five seeds; latency is the median.

Configuration	Accuracy (%)	Latency (ms)	Context
Dense	68.6 +/- 0.8	94	4k
Reranked	74.8 +/- 0.7	118	4k
No consistency term	72.9 +/- 0.9	117	4k

2.1. A closer look at 4k

Table 1 reports the complete measurements at the 4k budget. Reranking improves accuracy by 6.2 percentage points and increases median latency by 24 ms in this setup. Removing the consistency term reduces accuracy to 72.9%, while latency remains within 1 ms of the full pipeline. This ablation is consistent with a contribution from the training objective, although it does not isolate every possible interaction.

3. Discussion

The comparison suggests that a moderate context budget can retain much of the observed accuracy benefit while limiting latency. The 4k configuration reaches 74.8% accuracy with a median latency of 118 ms. The 8k configuration reaches 76.1%, but its additional context may contain redundant evidence. We therefore interpret the results as a trade-off within the evaluated setting, rather than as a universal choice of budget. Application requirements may favor a different point on this trade-off.

4. Scope and limitations

The corpus, questions, and measurements in this document are fictional and were created to demonstrate document layout. The evaluation design represents one language and one hardware setting. A real study would need independent datasets, uncertainty analysis, and measurements under deployment conditions before supporting broader claims. Layout changes in this example preserve the numerical values, methodological conditions, and interpretive limits of the text.